Master's Thesis

Multi Agent Reinforcement Learning-based Link Scheduling in NTN

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감사의 글

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Abstract

The development of sixth-generation (6G) networks aims to achieve global coverage through non-terrestrial networks (NTNs) utilizing low Earth orbit (LEO) satellites, with a focus on enhancing communication quality in high-traffic regions and areas lacking sufficient terrestrial infrastructure. However, the high altitude of LEO satellites increases the impact of signal fading, impairing accurate handover (HO) decisions by user equipment (UE). Additionally, the rapid mobility of the satellites results in frequent HOs, compromising network stability. This issue is particularly significant in Earth-moving cells (EMC) environments. Although conditional HO (CHO) and distance-based strategies have been introduced to address these challenges, they remain limited in effectively reducing unnecessary HOs (UHOs) and improving overall HO performance. To overcome these limitations, this thesis proposes an orbital information and distance (OID)-based CHO strategy. By using the positional relationship between the UE and satellite orbital information, the OID-based CHO enables more precise and efficient HO decisions, even in dynamic EMC environments where UE positions vary due to

cell movement. Simulation results demonstrate that the OID-based CHO significantly minimizes UHOs and reduces signaling overhead compared to conventional approaches. Furthermore, the proposed strategy can be implemented without requiring additional information exchange, offering a practical and efficient solution for NTN-based LEO satellite communication in 6G NTN systems.

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List of Abbreviations

LEO	Low Earth Orbit
UE	User Equipment
NTN	Non-Terrestrial Networks
TN	Terrestrial Networks
3GPP	3rd Generation Partnership Project
EMC	Earth-Moving Cell
CHO	Conditional Handover
BHO	Basic Handover
HO	Handover
MR	Measurement Report
SINR	Signal-to-Interference-plus-Noise Ratio
KPI	Key Performance Indicator
RSRP	Reference Signal Received Power
RLF	Radio Link Failure
UHO	Unnecessary Handover
HOPP	Ping-Pong Handover
ToS	Time of Stay

RB	Resource Block
RACH	Random Access Channel
OID	Orbital Information and Distance
FSPL	Free Space Path Loss
SIB	System Information Block
ISD	Inter-Site Distance
GPS	Global Positioning System
RRC	Radio Resource Control
ACK	Acknowledgment
TTT	Time-to-Trigger

Chapter 1. Introduction

1.1. Background and motivation

Low Earth Orbit (LEO) satellite communication has recently emerged as a promising next-generation technology for providing global coverage. The architecture of an LEO network consists of a multitude of satellites concurrently serving numerous terrestrial cells. However, this dense multi-satellite and multi-cell environment can lead to significant inter-cell interference. Furthermore, satellites require stable and high-capacity backhaul connections with gateways (GWs) to prevent data traffic bottlenecks and ensure seamless service for the cells they cover.

One of the most critical challenges in LEO satellite communications stems from the high mobility of the satellites. As satellites orbit the Earth at extremely high speeds, a ground terminal must frequently change its serving satellite. This high mobility necessitates frequent handovers and switchovers, which become a primary cause of degraded service continuity and can lead to temporary network disconnections.

In particular, conventional handover schemes based on received signal strength exhibit several limitations in this dynamic LEO environment.

Such schemes, which simply switch the connection to the satellite with the strongest signal, can trigger unnecessarily frequent handovers. Moreover, they often lead to the ping-pong effect, where handovers are rapidly repeated back and forth between two satellites at the cell edge. Consequently, the terminal's Time-of-Stay (ToS) with a single satellite becomes very short, which significantly degrades overall network efficiency and stability.

Therefore, to guarantee stable Quality of Service (QoS) and service continuity in LEO satellite networks, it is essential to develop intelligent handover and switchover schemes that can effectively manage the challenges posed by satellite mobility. This paper proposes a novel approach to address these pressing issues.

1.2. Contributions

To address the challenges of high mobility and dynamic network topology in LEO satellite networks, this paper proposes a novel handover and resource management scheme based on multi-agent reinforcement learning. The key contributions of this work are summarized as follows:

- **A Multi-Agent Reinforcement Learning (MARL) Framework for Joint Optimization:** We design a MARL framework where each satellite acts as an autonomous agent. These agents learn collaboratively to find the optimal policy for selecting both user cells and gateways simultaneously, adapting to the highly dynamic network environment without the need for a centralized con-

troller.

- **User Link Stability by Minimizing Handovers while Maximizing SINR:** Our proposed scheme enables the satellite agents to select user cells with a sophisticated dual objective. It aims to **maximize the Signal-to-Interference-plus-Noise Ratio (SINR)** for all served cells while concurrently **minimizing the total number of handovers**. This approach directly mitigates the ping-pong effect and enhances service stability for ground users.

- **Feeder Link Reliability by Preventing Bottlenecks and Reducing Switchovers:** The agents intelligently select gateways to **prevent traffic bottlenecks** in the backhaul network, ensuring efficient data routing. Crucially, this selection is performed while also **minimizing the frequency of feeder link switchovers**, which enhances the stability and reliability of the satellite-to-ground connection.

The effectiveness of our proposed scheme is validated through extensive simulations, which demonstrate significant improvements in network stability and user Quality of Service (QoS) compared to conventional, signal-strength-based methods.

1.3. Overview

The remainder of this thesis is structured as follows: Sec. II discusses handover, feeder link switching, and related research in beam hopping

and resource allocation. Sec. III presents the proposed multi-agent reinforcement learning framework in detail. Sec. IV discusses the simulation results and compares the proposed strategy with conventional approaches. Finally, Sec. V concludes the thesis.

Chapter 2. Related works

2.1. LEO satellite HO and FLSO schemes

In NTN-based LEO satellite communication system, two primary beam steering approaches can be identified, as illustrated in Fig. 2.1 [1]. In this figure, (a) represents the quasi-fixed cell approach. This method involves the satellite performing beam steering to maintain a fixed service area over a particular region as the satellite moves. In this scenario, as long as the satellite remains above the minimum elevation angle, no additional HO is required within the area. However, maintaining a fixed service area as the satellite moves requires complex and costly beam steering mechanisms, which significantly increase the complexity and cost of satellite design. On the other hand, as shown in Fig. (b), the EMC approach does not involve beam adjustments, reducing the design complexity and cost. However, as the satellite and its service cells move, users on the ground experience frequent cell transitions, leading to a higher number of HOs and increased signaling overhead. This thesis focuses the EMC approach to address the challenges caused by frequent cell transitions and aiming to improve overall system performance and stability.

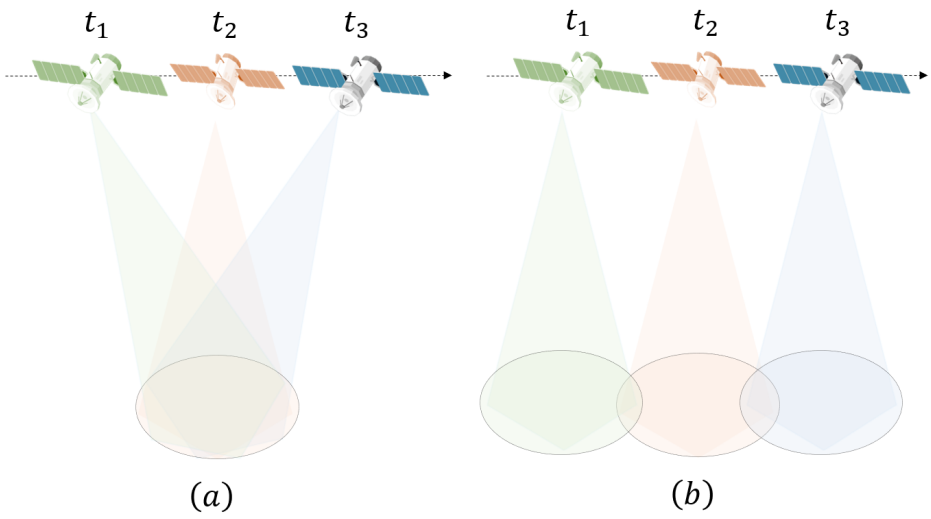


Fig. 2.1. Beam steering type: (a) quasi-fixed cell, (b) EMC

2.2. Reinforcement learning based Beam hopping and scheduling

Basic HO (BHO) is the traditional HO method where the serving cell initiates and controls the HO process [2]. First, when the HO conditions are satisfied for a certain period, known as the time-to-trigger (TTT), the UE sends a measurement report (MR) to the serving cell. Upon receiving the MR, the serving cell evaluates the measurement results and selects the most suitable target cell for the UE, making the HO decision. The serving cell then sends a HO request message to the target cell over the Xn interface to check if HO is possible. After receiving the HO Request acknowledgment (ACK) message from the target cell, the serving cell sends the HO command to the UE. Upon receiving the HO command, the UE immediately disconnects from the serving cell and attempts random

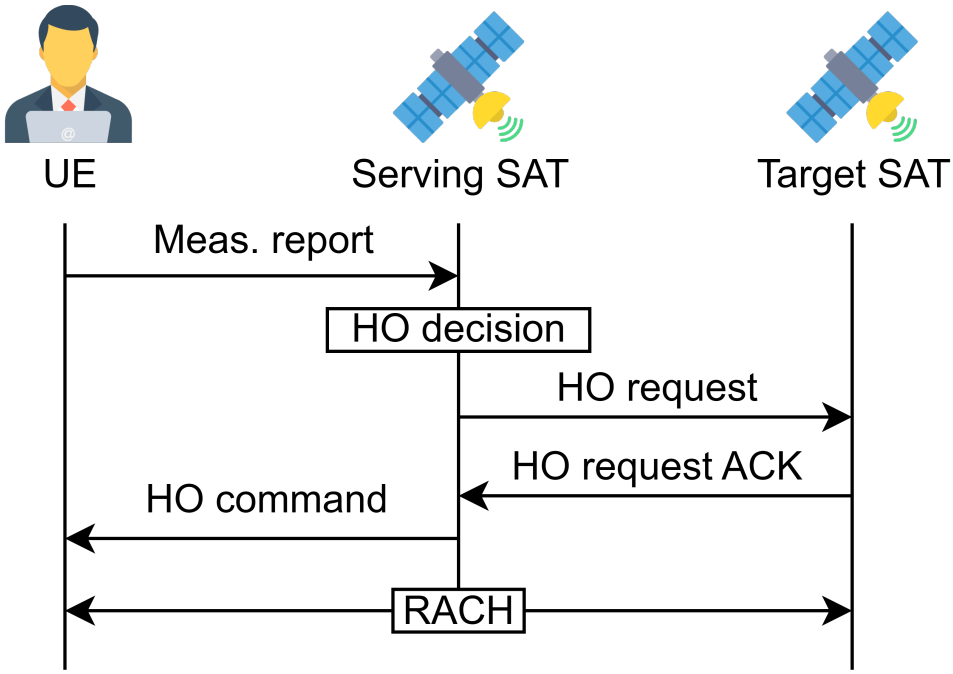


Fig. 2.2. Basic HO procedure

access (RA) with the selected target cell. The entire procedure is detailed in Fig. 2.2.

In TN, the cell radius is small due to the very small altitude compared to NTNs, so the propagation delay between the UE and the base station is minimized. However, in NTN, the significant distance between the UE and LEO satellites due to their high altitude causes much larger propagation delays compared to TN, introducing considerable latency in exchanging HO control messages between the UE and the base station [2]. This can lead to the selection of suboptimal target cells, as the MR used to make HO decisions are based on the signal conditions from previous moments. This issue is made worse by the dynamic environment of

LEO satellite cells, where the rapid movement of satellites makes precise timing for HOs important to prevent UHOs or RLF. This issue occurs because the target cell is selected based on MR results that are outdated and may no longer match the UE's optimal cell at the current location. Furthermore, due to the effects of signaling fading, there is a risk of selecting an incorrect target cell, leading to UHO or RLF. Therefore, employing the traditional HO method in NTN environments may degrade network performance.

To address the issues mentioned earlier, NTN adopts the CHO mechanism, initially proposed in NR [3]. CHO divides the HO process into two different phases: the preparation and the execution phase. Similar to the conventional method, when the HO condition is met, the UE sends an MR to the serving cell. However, in CHO, instead of selecting a single target cell, the serving cell prepares multiple candidate target cells based on the MR and initiates the HO request message procedure. After receiving the HO Request ACK message from these candidate cells, the serving cell sends the HO command to the UE. At this point, rather than immediately executing the HO, the UE evaluates the prepared candidate cells through an execution event and selects the optimal target cell for the HO. This approach enhances flexibility in target cell selection, especially in LEO satellite environments, helping to reduce UHO and RLF. The complete procedure of CHO is illustrated in Fig. 2.3.

This thesis proposes an OID-based CHO approach to enhance existing distance-based CHO mechanisms by incorporating satellite orbital information collected by the UE. Unlike previous methods that rely solely on

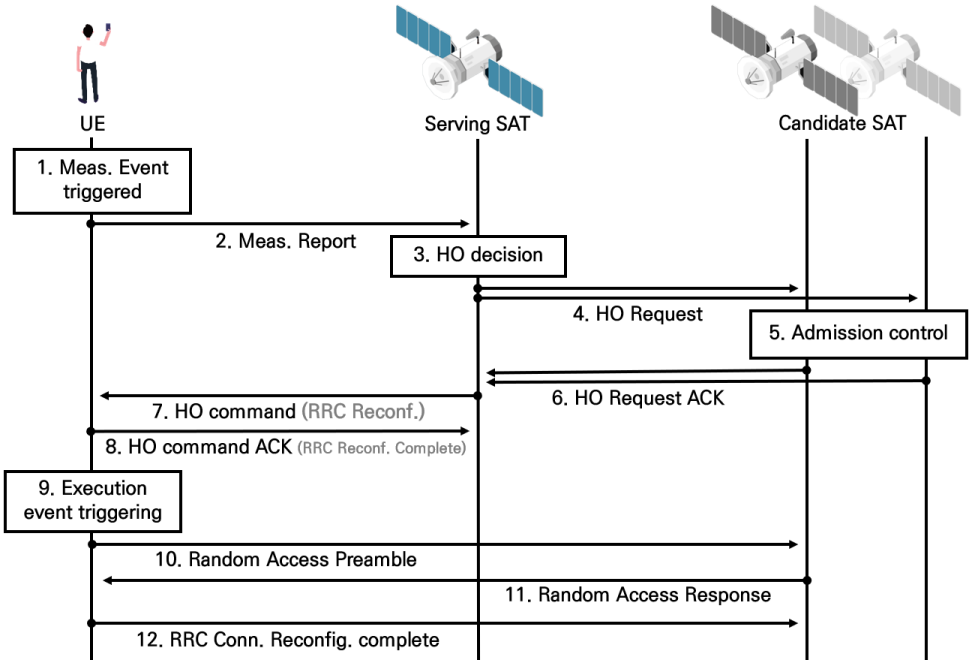


Fig. 2.3. Conditional HO procedure

distance or signal strength, the OID-based CHO considers the satellite's trajectory and predicted future positions, enabling more intelligent and efficient HO decisions. This approach addresses the limitations of existing methods and improves HO performance.

Chapter 3. Proposed multi-agent reinforcement learning-based link scheduling algorithm

3.1. System model

This thesis propose a beam-to-beam HO scenario in an environment with multiple satellites' spot beams in the same constellation, as shown in Fig. 3.1. The LEO satellite, equipped with several spot beams, operates at a low earth orbit and moves at 7.56 km/s. The HO is considered to occur only between satellites that move in parallel orbits within the same constellation. In the EMC environment, all cells use the same frequency, causing interference. The UE updates orbital information using system information block (SIB) 19 from the satellite and determines its position via global positioning system (GPS).

3.2. Problem formulation

To select an appropriate cell in an EMC environment and address the aforementioned issues, it is essential to define the positional relationship

between the UE and the cell. In this thesis, utilize the vertical distance from the cell trajectory to the UE, based on the position information at two instances along the cell trajectory, as illustrated in Fig. 3.2.

The X_p is the vertical distance between the cell trajectory and UE, calculated using the coordinates of both the satellite and UE. With the previous coordinates (x_{t0}, y_{t0}) and (x_{t1}, y_{t1}) provided to the UE through measurements or SIB 19, the trajectory is expressed as a linear equation:

$$(y_{t1} - y_{t0})x - (x_{t1} - x_{t0})y + (x_{t1}y_{t0} - y_{t1}x_{t0}) = 0. \tag{3.1}$$

Using the formula for the foot of a vertical line, we can calculate the vertical distance from the UE coordinates to the cell trajectory line.

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}. \tag{3.2}$$

Thus, the vertical distance X_p between the UE's position $(x_{t1}^{ue}, y_{t1}^{ue})$ and the cell trajectory line equation is defined as follows:

$$X_p = \frac{|(y_{t1} - y_{t0})x_{t1}^{ue} - (x_{t1} - x_{t0})y_{t1}^{ue} + (x_{t1}y_{t0} - y_{t1}x_{t0})|}{\sqrt{(y_{t1} - y_{t0})^2 + (x_{t1} - x_{t0})^2}}. \tag{3.3}$$

Through this approach, the X_p values for candidate cells can be calculated, thereby establishing the positional relationship between the UE and the cell trajectory in an EMC environment.

3.3. Proposed algorithm

The primary objective of the OID-based CHO is to reduce UHOs that occur in distance-based CHO methods. The proposed method encourages the UE to maintain its connection with the cell in the closest orbit by utilizing the previously defined vertical distance X_p. This strategy aims to select a cell that maximizes the Time of Stay (ToS) within the cell, thus providing an ideal scenario to minimize UHOs in the EMC environment. As shown in Fig. 3.3 (a), selecting the nearest cell in an EMC environment may lead the UE to HO from serving cell A to target cell B, and then back to cell C, which is in the same orbit as cell A. If the ToS in cell B is less than one second, this HO could be considered a UHO [4]. In contrast, as illustrated in Fig. 3.3 (b), selecting a cell with the same X_p value as the serving cell follows an ideal scenario where UHOs are prevented as the ToS increases. Additionally, if there is a cell in a closer orbit with a smaller X_p value than the serving cell, it may be more efficient to HO to that cell for better signal strength. Thus, this strategy involves deciding whether to maintain the current orbit or move to a closer orbit, providing flexibility in response to changes in the UE's location based on X_p values.

3.3.1 MDP modeling

The HO triggering condition of the proposed scheme follows the same distance-based entry condition specified in Section **??**. This mechanism consists of two thresholds, which the UE can obtain from the *distanceThresh* information in SIB 19 [5] of the satellite. However, the current design, in

which the serving satellite manages and transmits individual thresholds for each UE, may result in increased processing overhead. Additionally, conveying these thresholds to the UE solely through SIB 19 is practically limited. An alternative approach is to use Radio Resource Control (RRC) messages, but this also increases signaling overhead and may degrade HO performance.

In this thesis, a common threshold scheme is employed to address challenges associated with individual threshold configurations. This scheme is uniformly applied to all UEs in the serving cell to configure a threshold for triggering HO when approaching the cell edge. The proposed configuration is similar to conventional distance-based CHO mechanisms, but it utilizes cell edge information from both the serving and target cells to determine the thresholds. The first threshold, $Thresh_1$, determines whether the UE has exited the serving cell and incorporates the concept of an inner circle. As shown in Fig. 3.4, the inner circle utilizes inter-site distance (ISD) to defines the boundary of the serving cell that does not overlap with any candidate cell, and its radius, d_{rad}^{I}, is equal to $Thresh_1$. In cases where multiple d_{rad}^{I} values exist, the largest value is defined as the radius of the inner circle. This configuration ensures effective HO when the UE approaches the cell edge. The second threshold, $Thresh_2$, prevents HO from being triggered if the UE reaches the boundary of the serving cell but has not yet entered an appropriate range of the target cell. To achieve this, $Thresh_2$ is set within the outer circle of the target cell. This prevents unnecessary HOs and helps maintain a stable connection.

$$Thresh_1 = d^I_{rad} = ISD - d^T_{rad}. \tag{3.4}$$

The sizes of the inner and outer circles, which determine the threshold, can be set via parameters such as SIB 19, while the second threshold can be configured differently depending on system requirements. For instance, $Thresh_2$ can be set to trigger HO when the UE enters the valid signal strength range of the target cell, ensuring that the minimum signal strength of the target cell is met.

3.3.2 Proposed dueling DQN algorithm

When the UE meets the preparation event conditions, it sends a MR to the serving cell to initiate the HO procedure. The MR includes the UE's location information and the distance to each neighboring cell c_i, denoted as Ml_{c_i}. Upon receiving the MR from the UE, the serving cell determines a list of candidate cells. Traditionally, candidate cells are selected based on the MR information provided by the UE. However, in the proposed approach, the serving cell has access to the ephemeris information of the 1-tier cells, and thus can include all the surrounding cells of the UE as candidate cells. Additionally, neighboring cells that do not exceed the maximum candidate cell threshold d^c_{max}, based on the UE's location and Ml_{c_i} obtained from the MR, can also be included as candidate cells. Where, d^c_{max} represents the maximum distance considered for the surrounding cells, which can be determined as, for example, the maximum ISD that can be included as a 1-tier cell.

$$C_{candi} = \{c_i \in C \mid Ml_{c_i} \leq d^c_{max}\}. \tag{3.5}$$

The candidate cell set is defined as $C_{candi} = c_1, c_2, \ldots, c_n$. The proposed full OID-based CHO procedure is illustrated in Fig. 3.5, where the candidate cell selection described earlier corresponds to step 3, cell filtering. The subsequent procedures are identical to the CHO preparation phase in Section 2.2.2.

After receiving the HO command, the UE waits to verify the execution event. Upon satisfying the execution event, the UE executes an algorithm to select the target cell, C_t. This algorithm, based on the X_p values of the serving and candidate cells, is designed to prevent UHOs and ensure that the UE meets the HO conditions with the C_t.

First, for each cell c_n in C_{candi}, an $X_p^{c_n}$ value is calculated based on the UE's location. Similarly, the X_p value of the current serving cell, denoted as C_s, is represented as $X_p^{C_s}$, where C_s refers to the cell to which the UE is currently connected. To filter valid candidate cells for selection, the X_p values of candidate cells are compared with $X_p^{C_s}$:

$$C_{valid} = \{c_n \in C_{candi} \mid X_p^{c_n} \leq X_p^{C_s}\}. \tag{3.6}$$

If C_{valid} is formed, the UE selects the cell with the smallest X_p value in the C_{valid} set as the target cell C_t. However, if C_{valid} is empty, the UE waits until the next execution event is satisfied and then re-executes this process. To avoid indefinite waiting, a fallback mechanism is implemented. When the value of Ml_s exceeds a certain maximum threshold ($d^{c_s}_{max}$), a fallback mechanism is triggered. For example, it can mean the

distance at which the signal quality degrades below a predefined threshold. In such a case, the closest cell in the set C_{candi} is selected as C_t. This fallback mechanism is consistent with the conventional distance-based CHO scheme. Therefore, the overall C_t selection algorithm in the execution phase is structured as follows:

$$C_t = \begin{cases} \arg\min_{c_v \in C_{valid}} X_p^{c_v}, & \text{if } C_{valid} \neq \emptyset, \\ \arg\min_{c_n \in C_{candi}} Ml_n, & \text{if } C_{valid} = \emptyset \text{ and } Ml_s \geq d_{max}^{c_s}, \\ \text{Re-evaluate } C_{valid}, & \text{if } C_{valid} = \emptyset \text{ and } Ml_s < d_{max}^{c_s}. \end{cases} \tag{3.7}$$

This proposed algorithm effectively selects the optimal target cell for the UE in EMC environments by leveraging satellite orbital information and the UE's location. These inputs, already collected during distance-based CHO procedures, enable precise target cell selection with minimal processing overhead or additional signaling requirements. The pseudocode of this algorithm is detailed in Algorithm 3.1.

The proposed method is highly effective for selecting the optimal cell for the UE in an EMC environment. By utilizing satellite orbit information and the UE's location, which are already collected during distance-based CHO, the relative position to the orbit can be considered without additional information from the UE, enabling this process with minimal processing load or computational overhead.

Algorithm 3.1 OID-based CHO: Target Cell Selection

Input: $C_{\text{candi}} = \{c_1, c_2, \ldots, c_n\}$ (Candidate cells), C_s (Serving cell), $X_p^{C_s}$ (orbital distance of C_s)
Output: C_t (Target cell for HO) or `None`
if $C_{\text{candi}} = \emptyset$ **then**
    return `None` ▷ No candidate cells available
end if
if $Ml_s \geq d_{max}^{c_s}$ **then**
    $C_t \leftarrow \arg\min_{c_n \in C_{\text{candi}}} Ml_n$
    return C_t
end if
if Execution event is satisfied **then**
    Construct $C_{\text{valid}} \leftarrow \{c_n \in C_{\text{candi}} \mid X_p^{c_n} \leq X_p^{C_s}\}$
    if $C_{\text{valid}} = \emptyset$ **then**
        return `None` ▷ No valid cells, exit and retry execution event
    else
        $C_t \leftarrow \arg\min_{c_v \in C_{\text{valid}}} X_p^{c_v}$
        return C_t
    end if
end if
return `None` ▷ No valid cells selected after all conditions

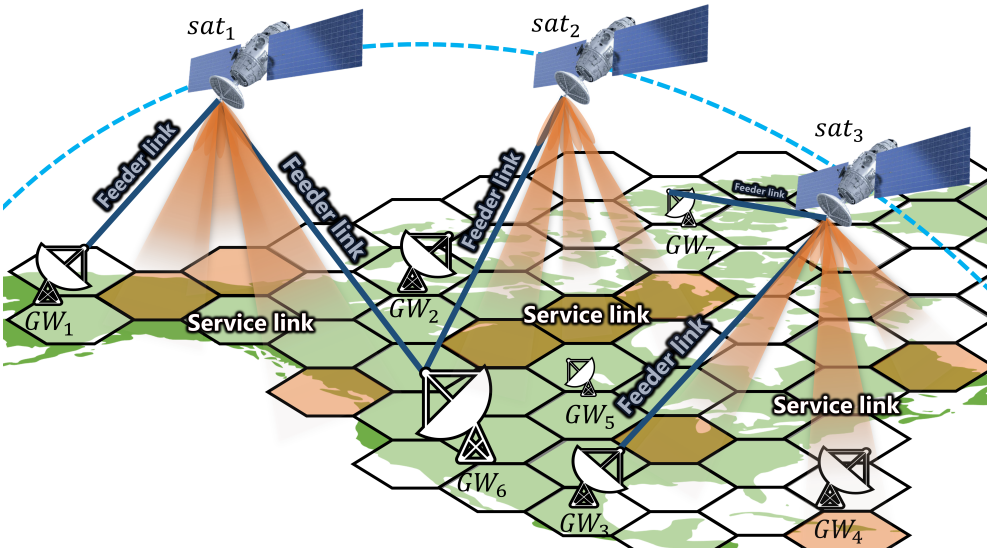


Fig. 3.1. HO and FLSO scenario: multiple satellites' beam to cell and GW

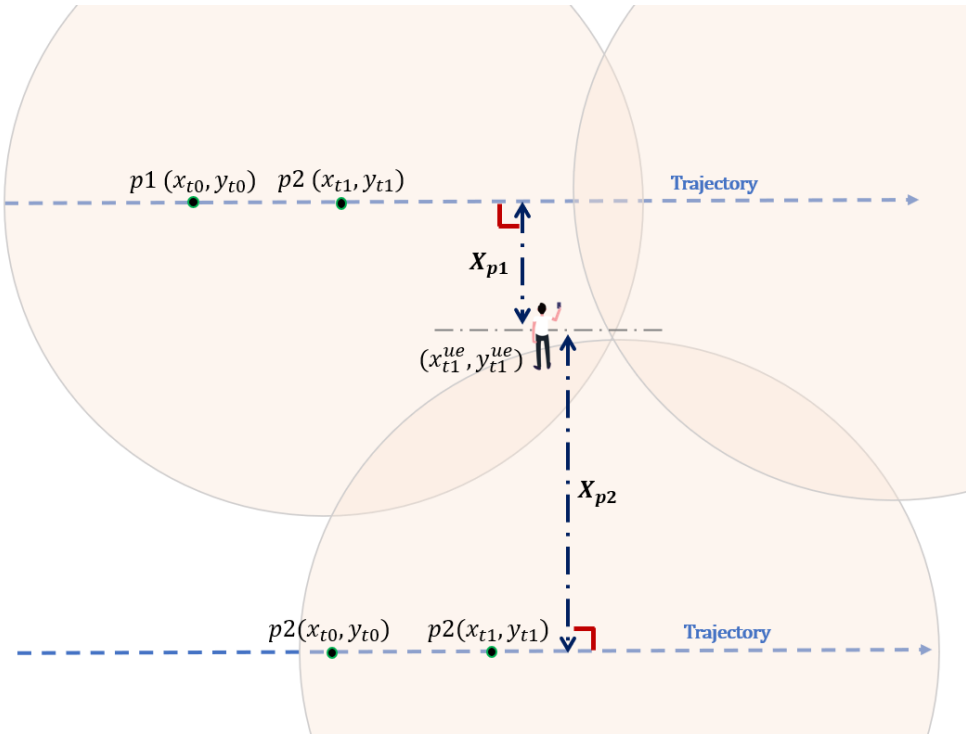


Fig. 3.2. Definition of UE's orbital distance (X_p)

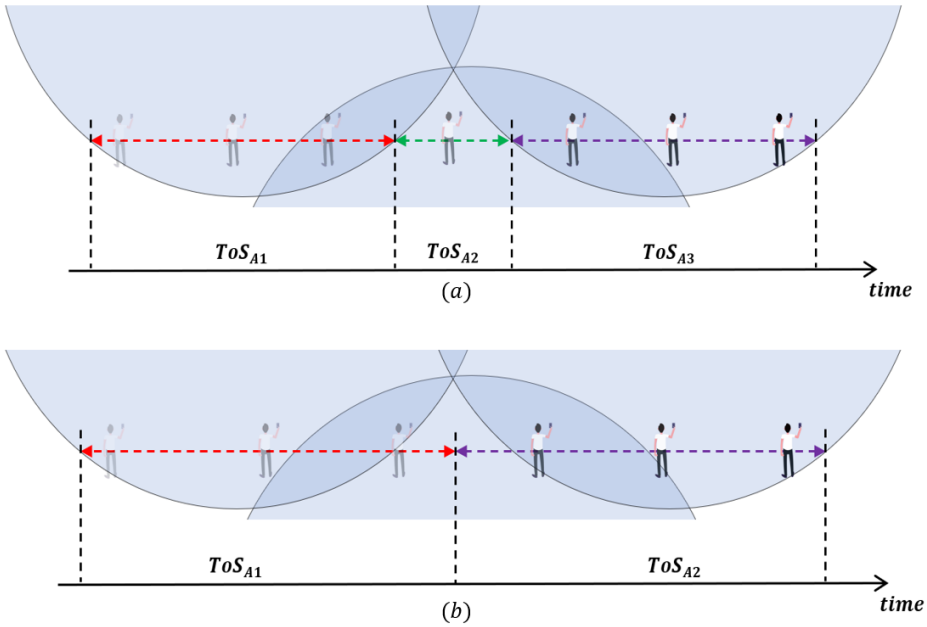


Fig. 3.3. Ideal scenario for UHO occurrence

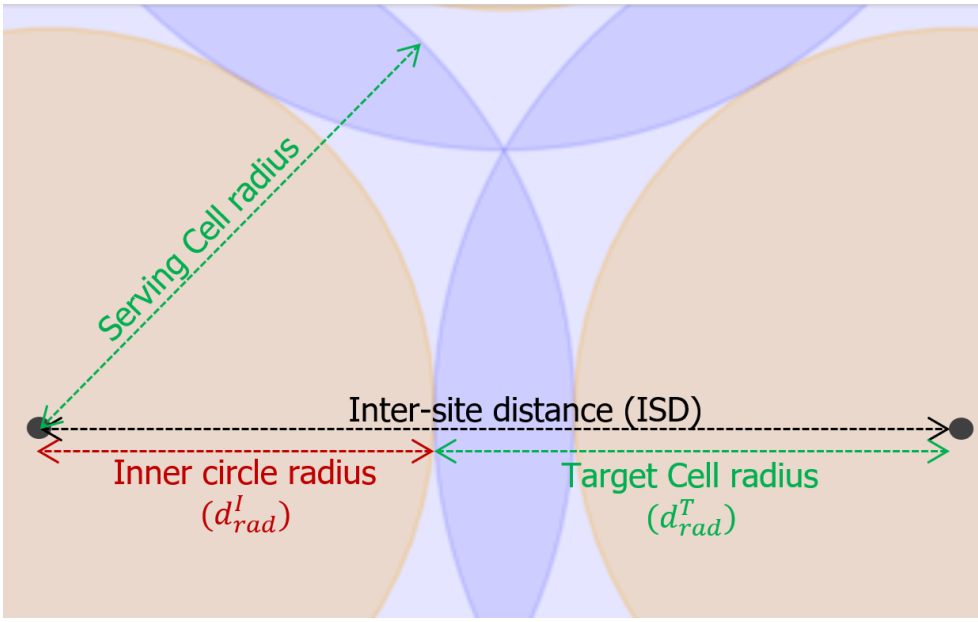


Fig. 3.4. Definition of inner-circle

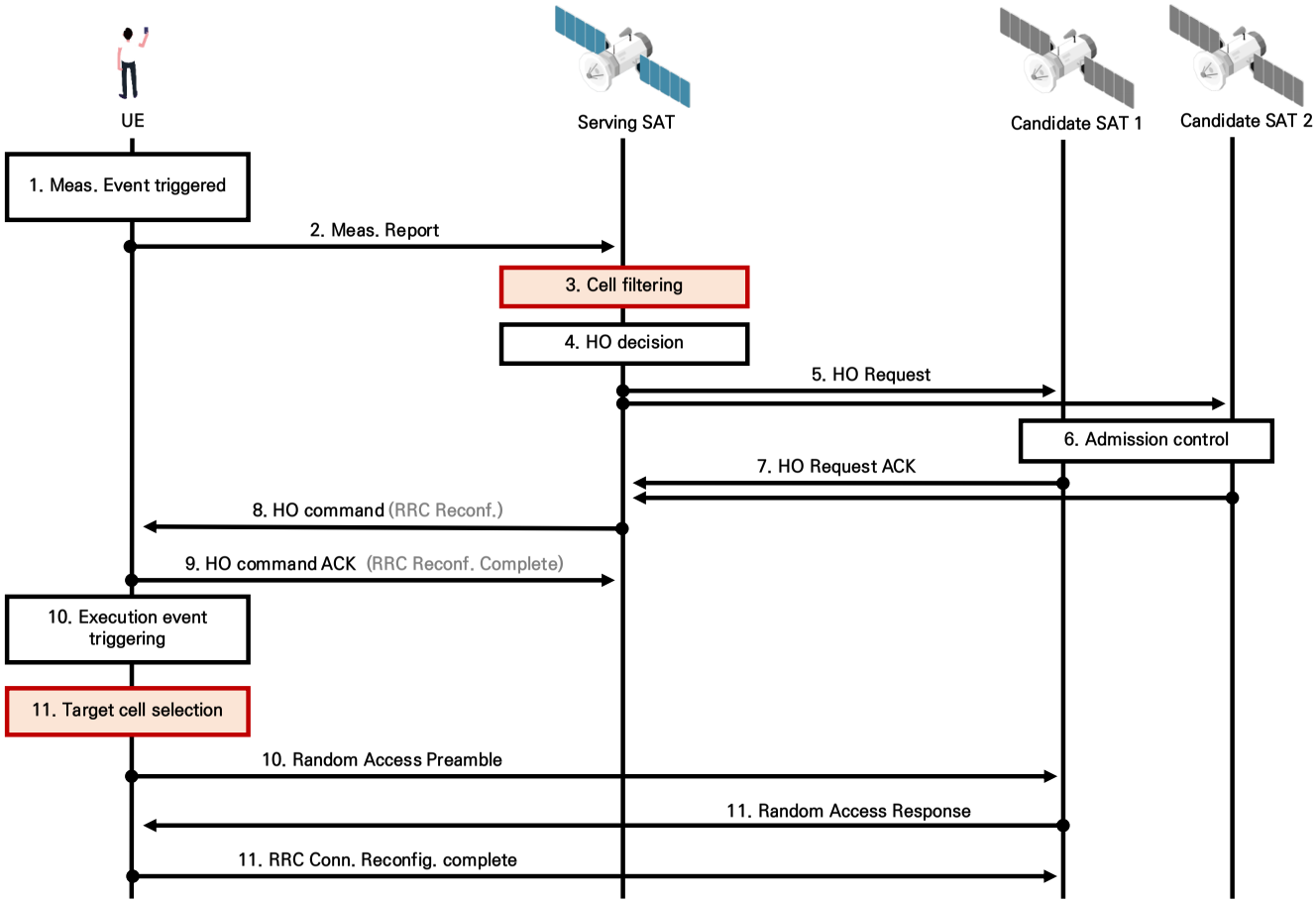


Fig. 3.5. Proposed OID-based CHO procedure

Chapter 4. Performance evaluation

4.1. Simulation setup

In this thesis, the case 9 scenario of the system-level simulation described in [1] is analyzed, with scenario D2 selected to account for the regenerative payload and EMC environment. The satellite antenna pattern is based on [6]. Each satellite has 19 slot-beams, and the ISD is arranged uniformly. Additionally, FR1 is considered, resulting in interference between spot-beams. In this simulation, In this simulation, 2,000 UEs are randomly distributed within the blue area of the satellite service region, as shown in Fig. 3.1. Since the satellite's speed is fast enough to ignore the UE's movement, a fixed UE is deployed. In addition, it is assumed that the UE can update its location information via the GPS. The detailed values are shown in Table. 4.1.

The path loss model for satellite communications is based on the large-scale satellite channel model provided in [1]. The total path loss between the UE and the LEO satellite is given by:

$$PL_{total}[dB] = PL_b + PL_{att} + PL_{gas} + PL_{entry}, \tag{4.1}$$

Table 4.1. System-level simulation: parameters for study case 9

LEO Satellite		**UE (Handheld)**	
Parameter	**Value**	**Parameter**	**Value**
Altitude	600 km	**Type**	Handheld
Antenna aperture	2 m	**Distribution**	100% outdoor
EIRP density	34 dBW/MHz	**Frequency band**	S-band (2 GHz)
3dB beamwidth	4.4127 deg	**FR factor**	Option: 1
Tx max gain	30 dBi	**Rx antenna gain**	0 dBi

where PL_b represents the basic path loss, PL_{att} accounts for atmospheric or ionospheric attenuation, PL_{gas} represents gas attenuation, and PL_{entry} refers to building loss. The basic path loss is defined as:

$$PL_b[dB] = FSPL(f_c, d) + SL(\alpha, f_c) + CL(\alpha, f_c), \tag{4.2}$$

where $FSPL(f_c, d)$ represents FSPL, d is distance between the satellite and UE, and c is speed of light. $CL(\alpha, f_c)$ represents clutter loss, and $SL(\alpha, f_c)$ refers to shadow fading modeled as $SL \sim N(0, \sigma_{SL}^2)$. The LOS probability, along with the values for $SL(\alpha, f_c)$ and $CL(\alpha, f_c)$, as shown in Table. 4.2, are provided in [6]. Total path loss is calculated as follows:

$$PL_b = P_{LOS}(FSPL(f_c, d)+SF)+P_{NLOS}(FSPL(f_c, d)+SF+CL. \tag{4.3}$$

Since frequencies below 6 GHz are considered, atmospheric and gas attenuation (PL_{att} and PL_{gas}) can be neglected [6], as can building loss PL_{entry} for outdoor users. This thesis focuses on dense urban regions in the 2 GHz band, so these attenuations are excluded.

Table 4.2. LOS Probability, SR and CL (S-band, dense urban)

Elev. angle (α)	**LOS prob.**	σ_{SF}**(LOS)**	σ_{SF}**(NLOS)**	**CL**
10°	28.2%	1.79 dB	8.93 dB	19.52 dB
20°	33.1%	1.14 dB	9.08 dB	18.17 dB
30°	39.8%	1.14 dB	8.78 dB	18.42 dB
40°	46.8%	0.92 dB	10.25 dB	18.28 dB
50°	53.7%	1.42 dB	10.56 dB	18.63 dB
60°	61.2%	1.56 dB	10.74 dB	17.68 dB
70°	73.8%	0.85 dB	10.17 dB	16.50 dB
80°	82.0%	0.72 dB	11.52 dB	16.30 dB
90°	98.1%	0.72 dB	11.52 dB	16.30 dB

The simulation set used in this study consists of eight different configurations designed to evaluate various HO strategies. These strategies are divided into signal strength-based CHO strategies using the A3 event and distance-based CHO strategies. Sets 1 to 4 focus on signal strength-based CHO strategies, considering CHO in satellite environments, and each simulation set is differentiated by the assigned offset and TTT values [7], [8]. The distance-based CHO strategies consist of three standardized sets defined by 3GPP. Sets 5, 6, and 7 implement the 3GPP-standardized distance-based CHO strategies [5]. Each set modifies $Thresh_1$, the threshold for exiting the serving cell, to compare performance variations based on changes in HO triggering points. In Set 5, HO is triggered when the UE exits the inner radius of the serving cell, initiating an early HO within the serving cell. Set 6 uses half of the ISD as the threshold, which corre-

Table 4.3. Simulation sets

#	**Strategy**	**Offset**	$Thresh_1$	$Thresh_2$	**TTT**
1	CHO (A3)	0 dB	-	-	0 ms
2	CHO (A3)	2 dB	-	-	0 ms
3	CHO (A3)	0 dB	-	-	256 ms
4	CHO (A3)	2 dB	-	-	256 ms
5	DCHO [5]	-	16.93 km (d^I_{rad})	23.12 km (d^T_{rad})	-
6	DCHO [5]	-	20.02 km ($\frac{ISD}{2}$)	23.12 km (d^T_{rad})	-
7	DCHO [5]	-	23.12 km (d^S_{rad})	23.12 km (d^T_{rad})	-
8	OID-based CHO	-	16.93 km (d^I_{rad})	23.12 km (d^T_{rad})	-

sponds to the moment when the target cell becomes closer to the UE than the serving cell. Set 7 allows HO to be triggered when the UE reaches the outer radius of the serving cell, i.e. the maximum boundary of the cell, while $Thresh_2$ is satisfied. Each set is designed to observe performance variations based on the UE's HO triggering location. The final set, Set 8, evaluates the newly proposed distance-based CHO mechanism, the OID-based CHO. Additionally, the K filter, which represents the number of averaged measurements, is considered as 1 for all simulation sets.

4.2. Key Performance Indicators

The first is the average downlink (DL) signal-to-interference-plus-noise ratio (SINR), which represents the quality of the signal received by the UE from the serving cell. It measures the signal strength from the serv-

ing cell relative to the interference and noise from adjacent cells. Higher SINR implies better signal quality and more stable communication, significantly enhancing network performance, data transmission rates, and communication reliability. In this thesis, the average SINR values of each UE were analyzed during the simulation to assess how effectively the network is operating.

RLF occurs when the UE's SINR falls below the threshold Q_{out} (typically -8 dB) and does not recover within a certain time (T310 timer, typically 1 second). If the SINR recovers above the Q_{in} threshold (typically -6 dB), the timer stops, and no RLF occurs. RLF indicates a breakdown in communication, representing a critical issue when the network is unable to maintain stable connections. In this thesis, the number of RLF occurrences was measured for each UE and calculated as the rate of RLF per second per UE to evaluate HO performance. The higher the number of RLF occurrences, the lower the network stability and the higher the likelihood of HO failures. More detail can be found in [3], [5] and [7].

ToS measures the duration the UE remains connected to a specific cell, indicating how long the connection is maintained in that cell [7]. In this thesis, ToS is defined as the time from when the UE sends the HO complete message to cell A until it sends a HO complete message to cell B. A shortToS implies frequent HOs, which can lead to unnecessary resource wastage and degrade communication quality.

UHO is defined as when the UE performs a HO from the serving cell to a target cell, only to HO back to the original serving cell or another cell within 1 second [4]. As shown in Fig. 4.1(a), UHO is a key metric for

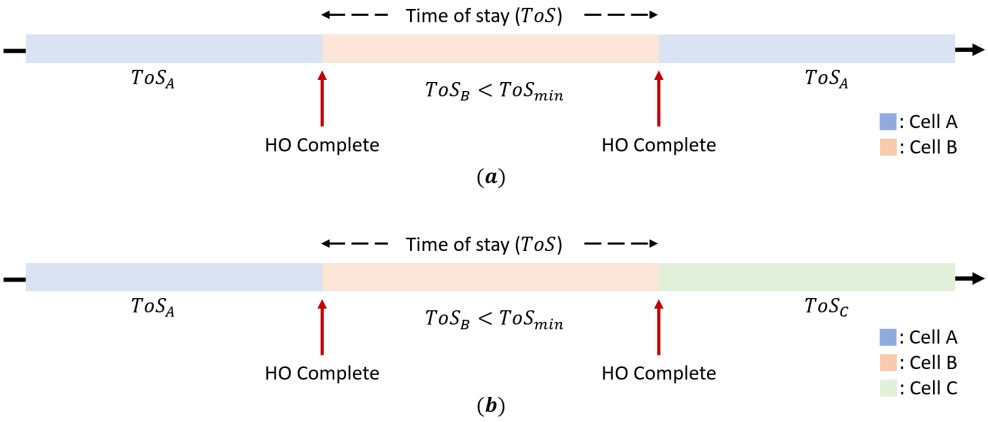


Fig. 4.1. Definition of UHO and HOPP according to ToS

evaluating the efficiency of the HO algorithm. A high UHO occurrence rate indicates an increase in unnecessary signaling overhead and inefficient HO decisions. In this thesis, the UHO occurrence rate was measured based on the number of UHO events during the simulation. A low UHO rate indicates more accurate HO decisions and efficient use of system resources. HO Ping-pong (HOPP), shown in Fig. 4.1(b), is a specific type of UHO where the UE returns to the original serving cell within 1 second [9] and [10]. This represents an even more inefficient case of UHO, leading to excessive signaling overhead and instability in the connection. This thesis evaluates the performance of each strategy by calculating the occurrence ratio of HOPP and UHO among the total number of HOs.

The average number of resource blocks (RBs) used during the HO process is an important metric for evaluating signaling overhead and resource efficiency. RB usage is computed using the RBs consumed for each step of the HO process, which allows us to estimate the signaling

Table 4.4. Number of RBs in the HO control message used

Type	Description	RBs
N_{MR}	Resource blocks used for the MR	1
N_{cmd}	Resource blocks used for the HO command	2
N_{RA}	Resource blocks used for RA	6
N_{cnf}	Resource blocks used for HO confirmation	1

overhead of the system. The RBs generated during the HO process are referenced in [11] and listed in Table 4.4.

4.3. Results and Discussion

Fig. 4.2 clearly illustrates the RLF rate per UE per second across various HO strategies. The blue bar graph represents the results of signal-based CHO, which evaluates the signal strength of the serving and candidate cells. However, due to environmental factors, weakening, and fading, the signal strength does not increase or decrease steadily, leading to a higher likelihood of additional HOs occurring within the cell or near its boundaries. The RLF rate in Sets 3 and 4 is higher than in Sets 1 and 2. This indicates that adjusting HO parameters such as signal strength offset and TTT to delay HO decisions can allow for more careful HOs and help resolve such issues. However, increasing parameter values also makes the HO conditions more difficult to satisfy, causing delays in HO execution. In such cases, weakening of the signal strength in the serving cell leads to a reduction in DL SINR, increasing the risk of RLF. On

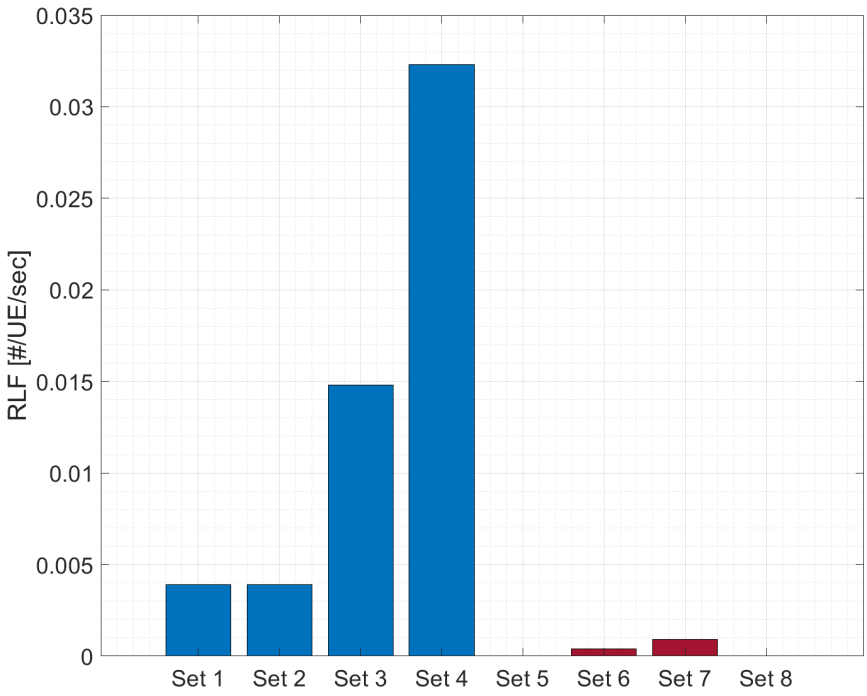


Fig. 4.2. RLF performance results

the other hand, distance-based CHO strategies are unaffected by signal strength variability and can perform HOs at optimal cell and UE locations. Consequently, Sets 5 to 7 exhibit significantly lower RLF rates compared to Sets 1 to 4. However, the RLF gradually increases in Sets 6 and 7 compared to Set 5. This is because, in certain sets, the event satisfaction point is farther from the serving cell, resulting in RLFs primarily caused by weakening of the signal strength in the serving cell. This delay in HO timing can lead to frequent HOs near the serving cell boundary. In contrast, the proposed method evaluates target cell selection to delay HO

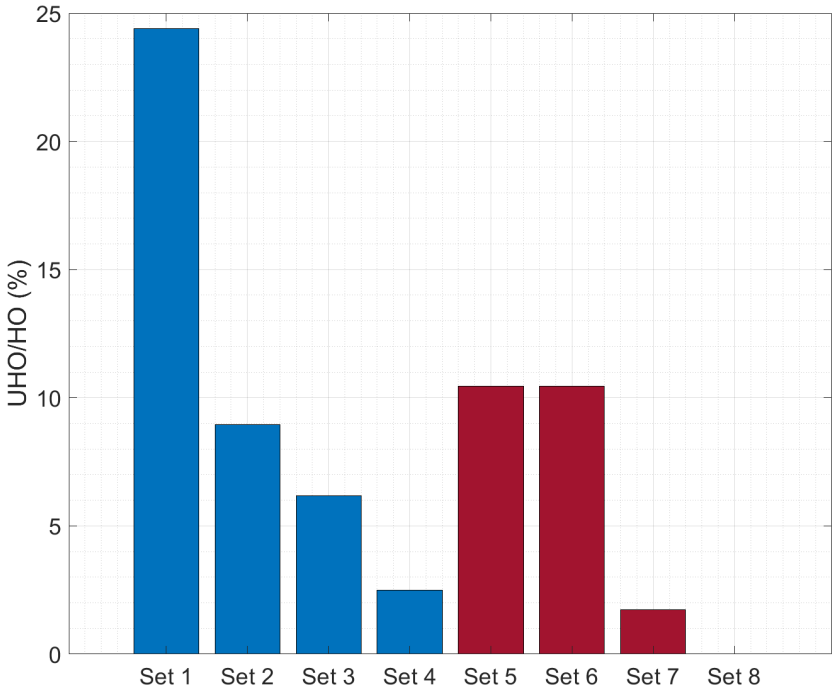


Fig. 4.3. UHO performance results

timing when necessary. Unlike Sets 6 and 7, it introduces an exception condition that triggers HO to the nearest cell when the signal strength of the serving cell drops below a specific threshold, even if HO timing is delayed due to target cell selection. As a result, the proposed method effectively prevents RLF caused by weakening of the signal strength, showing its effectiveness in reducing RLF occurrences.

Fig. 4.3 and Fig. 4.4 show the ratio of UHO and HOPP among the HOs that occurred in the UE during the simulation, respectively. In signal strength-based CHO using the A3 event, Sets 1 to 4 exhibit a significant

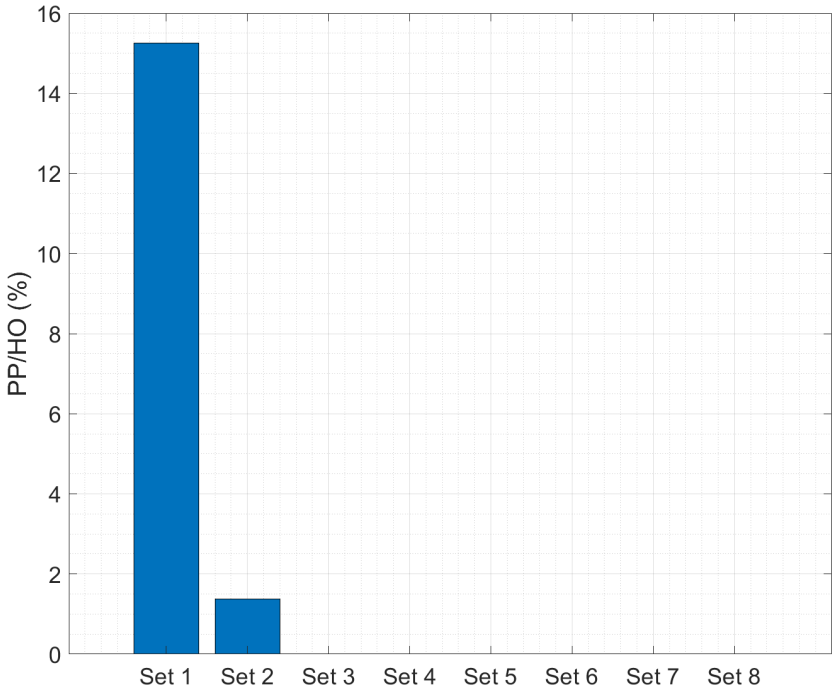


Fig. 4.4. HOPP performance results

decrease in the UHO occurrence rate as the offset and TTT values are increased. In Set 1, frequent HO attempts occur due to the small difference between the center and boundary of the cell and the effects of signal strength fading, resulting in a relatively high UHO rate and frequent HOPP occurrences due to incorrect triggers. In most cases, this is attributed to UHOs that return to the previous serving cell due to incorrect triggers. However, increasing the offset and TTT reduces the number of incorrect HO triggers caused by signal strength fading, leading to a decrease in both UHO and HOPP occurrences. This indicates that

delaying HO through offset and TTT settings reduces the frequency of HO attempts compared to immediate HO execution. Furthermore, as signal strength attenuation in the serving cell becomes more pronounced, the total number of HOs, UHOs, and HOPPs also decreases. On the other hand, in Sets 5 to 7, which use distance-based CHO conditions, the UHO occurrence rate is significantly lower than in Set 1. This is because distance-based CHO is unaffected by signal strength fading and provides a more reliable HO trigger based on linear changes in distance information within the EMC environment. In particular, in Set 7, where the HO timing is delayed closer to the boundary of the serving cell, stricter HO conditions further reduce the UHO rate. However, as shown in Fig. **??**, UHOs can still occur in distance-based CHO when the UE deviates from its trajectory and meets the HO condition for an adjacent orbital cell. Therefore, while HOPP occurrences can be significantly decreased, it is not possible to completely eliminate UHO occurrences. Additionally, delaying HO timing by increasing the offset can impact the average DL SINR due to signal strength attenuation. However, in the case of the proposed OID-based CHO strategy in Set 7, it was confirmed that UHO occurrences can be effectively decreased by connecting the UE to the most suitable cell based on its position using orbital information.

Fig. 4.5 illustrates the average DL SINR experienced by the UE during the simulation as a box plot. As shown in the plot, Set 1, which uses signal strength-based CHO without any HO parameter settings, is more likely to select an incorrect target cell compared to distance-based CHO strategies due to the effect of signal strength fading, resulting in an over-

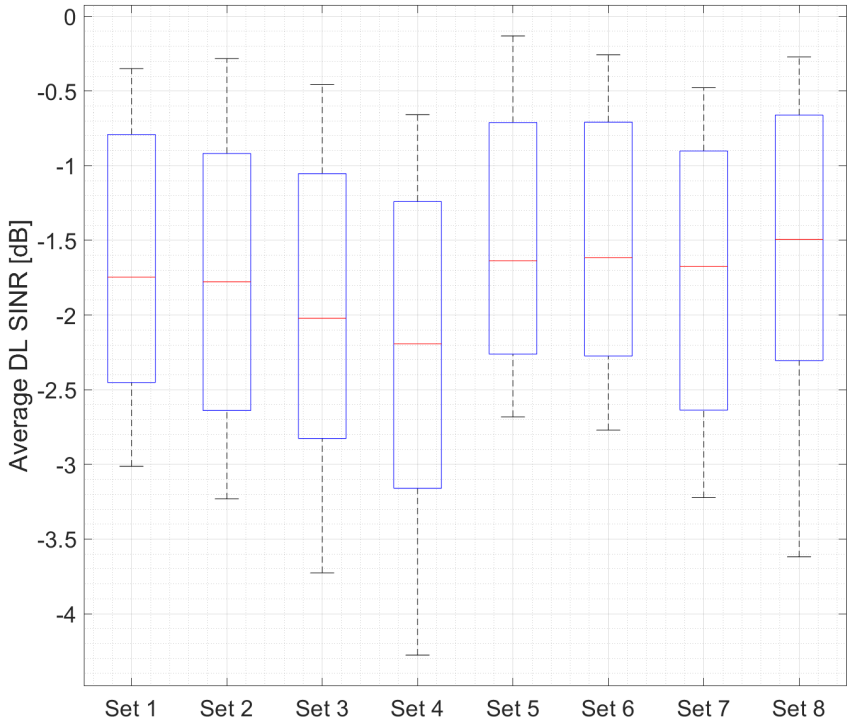


Fig. 4.5. Box plot of DL SINR performance results

all lower signal strength distribution. Additionally, Sets 2, 3, and 4, which apply HO parameters, exhibit progressively lower average DL SINR distributions compared to Set 1, which immediately performs HO to the cell with higher signal strength. The signal strength-based CHO method cannot completely avoid the effects of signal strength fading and faces challenges in selecting ideal cells, often requiring acceptance of signal attenuation. Furthermore, signal strength-based strategies tend to have a relatively high risk of RLF occurrence, further lowering the average signal

strength. In contrast, distance-based CHO shows higher average signal strength compared to the signal strength-based CHO. This is because it avoids the effects of signal strength fading and improves DL SINR performance by utilizing clear distance information to perform HO at an optimal location. Specifically, strategies such as those in Sets 4 and 5, which execute HO immediately upon entering the valid region of the target cell from the serving cell's area, achieve the highest average signal strength. However, in Set 6, where HO timing is significantly delayed, the average DL SINR may decrease. This suggests that additional processes are needed to adjust appropriate thresholds based on the situation when relying on HO parameters. The proposed OID-based CHO provides flexible timing for target cell selection, allowing HO to be delayed when a more suitable candidate cell is expected, even if a closer cell is available. As a result, it exhibits a wider range of average DL SINR distributions compared to conventional distance-based CHO strategies. The proposed method demonstrates its ability to select the most suitable cell for the UE state in an EMC environment and shows that it can enhance overall DL SINR performance without relying on specific HO offset adjustments.

These results can be confirmed in Fig. 4.6, which illustrates the CDF (cumulative distribution function) of all DL SINR values observed during the simulation period. Among all strategies, Set 1 shows the highest performance among signal strength-based strategies but is lower than Sets 5 and 6, which exhibit the best DL SINR trends. In contrast, Sets 2, 3, and 4, which applied HO offsets, show the lowest performance in the DL SINR distribution. This indicates that signal strength-based strategies

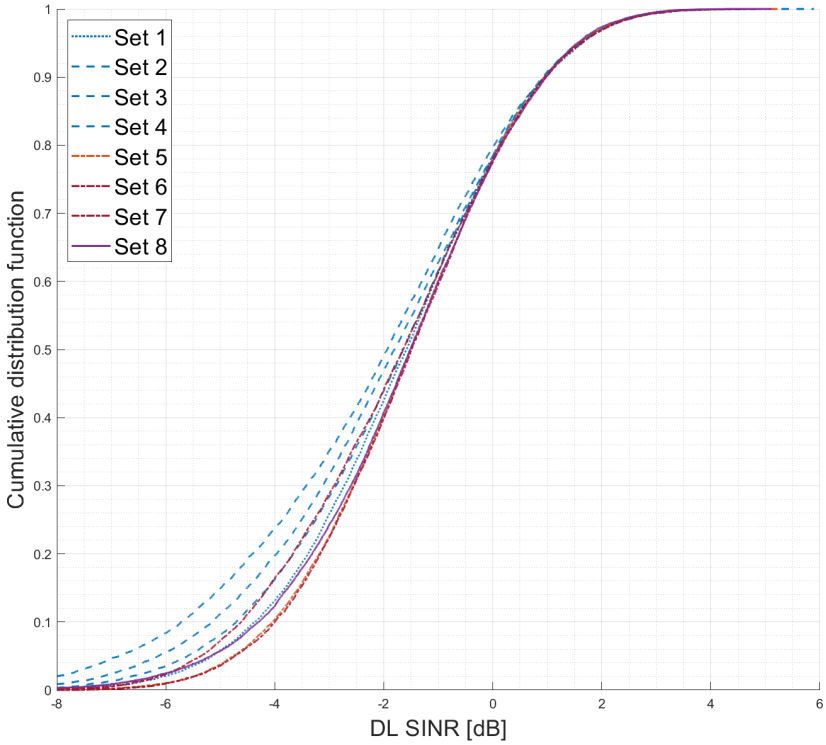


Fig. 4.6. CDF of DL SINR performance results

face challenges in selecting and maintaining the ideal cell in the NTN EMC environment. On the other hand, Sets 5 and 6 demonstrate the highest signal strength distribution by immediately selecting the most ideal cell. However, Set 7 exhibits a lower signal strength distribution than Set 1 due to signal attenuation in the serving cell caused by delayed HO timing through threshold adjustments. This highlights that even in distance-based CHO strategies, threshold adjustments significantly impact performance. Finally, the proposed strategy in Set 8 shows slightly

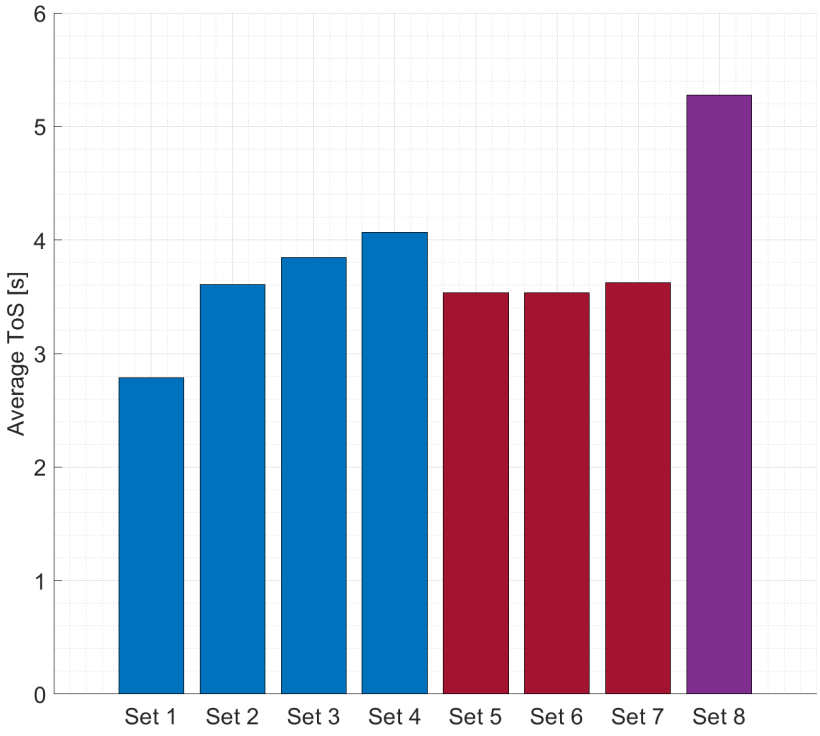


Fig. 4.7. ToS performance results

lower DL SINR tendencies compared to Sets 5 and 6 but outperforms signal strength-based strategies. Notably, it demonstrates a higher DL SINR performance distribution compared to Set 7, where UHO was reduced by adjusting the distance threshold. Therefore, when analyzed in relation to the previous UHO results, it is confirmed that the proposed strategy can achieve the most optimal performance by appropriately balancing the trade-off between DL SINR and UHO through threshold adjustments in distance-based CHO.

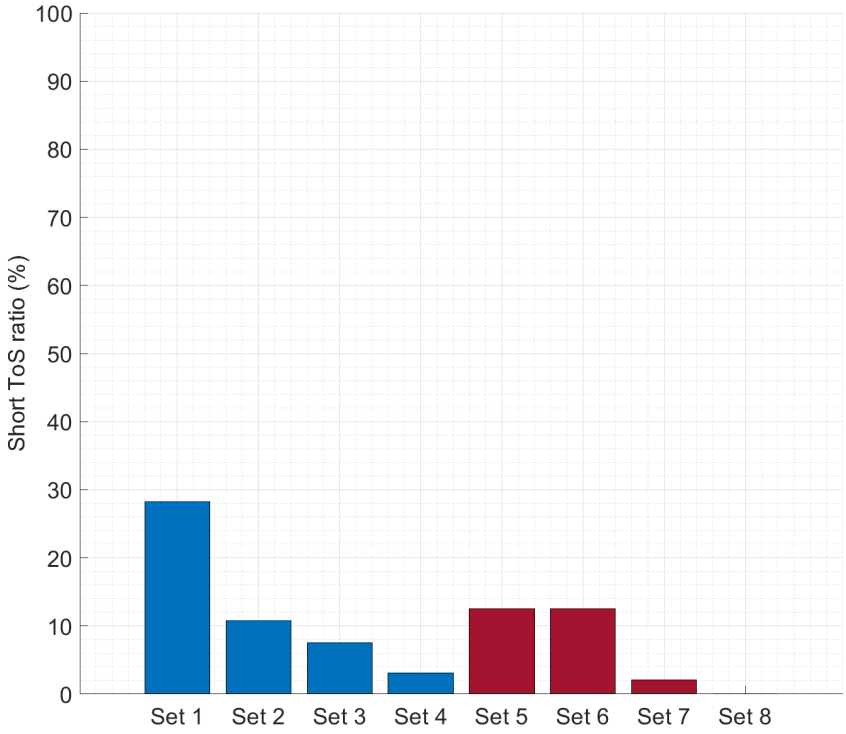


Fig. 4.8. ShortToS performance results

Fig. 4.7 illustrates the average ToS experienced by the UE during the simulation, while Fig. 4.8 shows the ratio of shortToS to total ToS. Here, shortToS refers to ToS durations of less than 1 second. Set 1 recorded the highest UHO and HOPP occurrence rates among all strategies, resulting in the lowest ToS and the highest shortToS ratio. This is because HO conditions are frequently satisfied in this strategy, leading to continuous UHO, especially HOPP, which significantly degrades service quality due to frequent service interruptions during the HO process. In contrast, Sets

2, 3, and 4 showed an improving trend in ToS performance as HO parameters increased in the signal strength-based method, with noticeable improvements in the shortToS ratio. Additionally, as HOPP occurrences decreased, overall ToS increased. Sets 5, 6, and 7 applied distance-based methods to perform HOs at ideal locations, resulting in higher ToS compared to Set 1, with no HOPP occurrences. However, due to UHO, the average performance of ToS was lower than in Sets 2, 3, and 4, and the shortToS ratio was relatively higher. In particular, sets 5 and 6 showed some improvements in set 6, where threshold adjustments were made to modify the HO timing near the cell boundary. However, the impact of small positional changes in boundary areas was relatively limited compared to the overall cell size, resulting in minimal improvement in ToS performance. On the other hand, the proposed method, Set 8, recorded the highest ToS among all comparison groups by selecting the appropriate target cell based on the cell trajectory and UE location.

Finally, Fig. 4.9 measures the average RB usage for HO control messages to evaluate how much signaling overhead within the network has been reduced. Set 1 recorded the highest UHO and HOPP occurrence rates among all strategies, leading to the highest number of HO attempts and, consequently, the highest average RB usage. In contrast, Sets 2, 3, and 4, which applied HO parameters, made it more challenging to satisfy HO conditions compared to Set 1, resulting in fewer HO occurrences and a corresponding reduction in average RB usage within the network. In the case of Set 5, the average RB usage was lower than Set 1 but higher than Sets 2, 3, and 4, following a trend similar to UHO perfor-

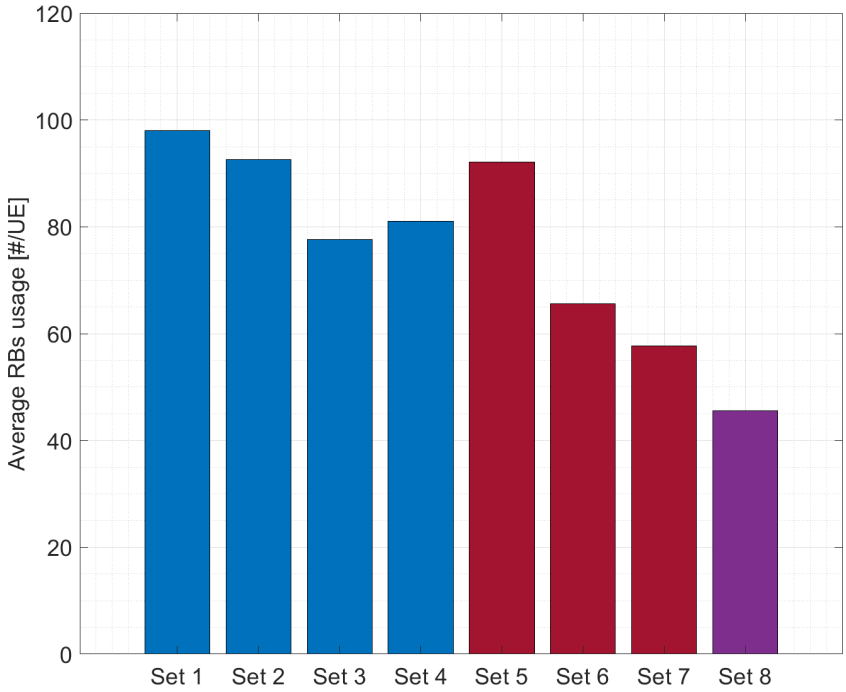


Fig. 4.9. Average RB usage results

mance. Meanwhile, Sets 6 and 7 showed significantly reduced RB usage compared to Sets 1 through 5, as stricter HO conditions reduced the number of HO attempts. But, the method proposed in this thesis, Set 8, was evaluated with stricter criteria for target cell selection and HO execution than other strategies. By allowing target cell selection and HO delays to be controlled based on conditions, the method can guide the UE to the most ideal cell, minimizing average RB usage. Therefore, it was confirmed that the proposed approach effectively addresses the issue of signaling overhead within the network, demonstrating its potential for

improvement in network efficiency.

Table 4.5 summarizes the detailed results of all simulations. In conclusion, the proposed OID-based CHO strategy effectively prevents UHO and HOPP occurrences while demonstrating performance improvements over existing HO strategies in key metrics, such as extended average ToS and minimized RB usage. These improvements have the potential to significantly enhance overall network performance in EMC-based LEO satellite communication systems. Moreover, the proposed method is not influenced by HO parameter settings. The key parameter, X_p, is utilized through distance-based CHO by leveraging the default position information, eliminating the need for additional data exchange. As a result, the method imposes minimal processing burden on the terminal, making it a practical and feasible approach for real-world deployments.

	Set 1	Set 2	Set 3	Set 4	Set 5	Set 6	Set 7	Set 8
RLF [#/UE/sec]	0.004	0.004	0.015	0.032	0	0	0.001	0
Avg UHO ratio [%]	24.387	8.95	6.182	2.495	10.438	10.438	1.736	0
Avg HOPP ratio [%]	15.259	1.377	0	0	0	0	0	0
Avg DL SINR (25%) [dB]	-2.457	-2.652	-2.844	-3.161	-2.266	-2.284	-2.647	-2.315
Avg DL SINR (50%) [dB]	-1.657	-1.791	-1.968	-2.205	-1.533	-1.507	-1.754	-1.612
Avg DL SINR (75%) [dB]	-0.806	-0.922	-1.067	-1.257	-0.722	-0.710	-0.902	-0.673
Avg ToS [sec]	2.786	3.609	3.846	4.067	3.534	3.534	3.626	5.276
ShortToS ratio [%]	28.233	10.811	7.556	3.088	12.551	12.551	2.101	0
Avg RBs usage [#/UE]	98	92.57	77.57	81	92.08	65.58	57.66	45.6

Table 4.5. Summary of Simulation Results by Set

Chapter 5. Conclusion

This thesis presents an OID-based CHO strategy designed to address mobility issues that still occur in distance-based CHO in LEO satellite EMC environments. Rapid satellite movement in EMC environments often leads to frequent HO, which causes issues such as UHO and RLF. The proposed OID-based CHO extends the existing distance-based strategy by newly defining the concept of distance from the orbit line, enabling more appropriate cell selection according to UE conditions. This allows the UE to remain connected to the most optimal satellite cell, providing a more stable communication environment even under rapid satellite movement. Simulation results show that OID-based CHO provides significant improvements over existing signal strength-based and distance-based strategies, including reduced RLF and UHO occurrence rates, and reduced RB usages. In addition, this strategy also effectively compromises the trade-off relationship between reduced UHO occurrence rates and reduced DL SINR in existing strategies by adjusting the offset, thereby achieving optimal results. Furthermore, the proposed approach reduces UE complexity and resolves practical challenges in parameter delivery by using a common offset for all UEs instead of assigning individual offsets.

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